

Jayana Estacio

jpstacio@uw.edu · linkedin.com/in/jpstacio · github.com/jpstacio

EDUCATION

University of Washington, Tacoma

Doctor of Philosophy in Computer Science and Systems

Master of Science in Computer Science and Systems — GPA: 3.82/4.0

Bachelor of Science in Computer Science and Systems

Expected Spring 2029

December 2026

June 2025

PUBLICATIONS & AWARDS

Temporal State-Conditioned Retrieval for Sequential Protocols: A BM25-Based Approach

2026

Estacio, J. and Teredesai, A. *Agent4IR Workshop at KDD 2026, accepted*. Jeju, South Korea.

Outstanding Graduate Research Award

2026

Annual Dean's Award, School of Engineering and Technology, University of Washington, Tacoma

RESEARCH EXPERIENCE

Research Assistant — Health News Evaluation

June 2026 – Present

NSF National Artificial Intelligence Research Resource Pilot | PI: Dr. Naeemul Hassan

- Building an evidence-grounded framework to evaluate health-news accuracy using a curated corpus that links peer-reviewed papers, institutional press releases, and resulting news articles at the claim level.
- Developing an annotation taxonomy for scientific accuracy, uncertainty, benefit-harm framing, and evidence support; evaluating LLM-assisted labels against expert annotations through agreement measurement and error analysis.
- Building reproducible Python pipelines to ingest, align, and version publication-press release-news triples, supporting iterative annotation and evaluation across the corpus.

Graduate Researcher — Information Retrieval Thesis

March 2026 – Present

University of Washington | Advisor: Dr. Ankur Teredesai

- Designed a temporal state-conditioning method for BM25 retrieval over sequential clinical protocols, combining query-side JSON-to-lexical linearization with a corpus-side hard phase filter.
- Developed a protocol-independent schema intended to transfer across sequential clinical workflows without requiring changes to the retrieval pipeline.
- Built a **278-event benchmark** from 261 r/IVF threads and a 52-passage evidence corpus spanning all nine IVF protocol phases, evaluated using nDCG, Recall, MRR, Phase-Valid Precision, and Wrong-Phase Rate.
- Improved nDCG@10 by **0.112** over stateless BM25; the hard phase constraint increased Phase-Valid Precision from **0.161 to 1.000** and eliminated wrong-phase retrievals. Conducted phase-leakage, corrupted-state, and partial-state ablations; work accepted to Agent4IR at KDD 2026.

Undergraduate Research Assistant

June 2024 – June 2025

University of Washington | Advisor: Dr. Juhua Hu

- Curated a **200,000+ study multimodal MIMIC-CXR dataset** and engineered an automated CheXpert and CheXbert labeling pipeline with distribution-level checks for label drift and data-quality failures.
- Analyzed LLM error patterns involving negation and diagnostic uncertainty, producing findings that informed the MIRAGE retrieval architecture.
- Collaborated with clinical and machine-learning researchers to compare automated labels with radiologist annotations, document pipeline decisions, and support reproducible downstream experiments.
- Delivered a guest lecture on the project to undergraduate and graduate computer science students.

SERVICE & LEADERSHIP

Director of Student Technology

June 2023 – June 2024

University of Washington, Tacoma

- Directed an **8-member committee** that conducted the university's first campus-wide technology survey, collecting more than **200 responses**.
- Led the committee's first bylaws revision in **seven years** and presented survey findings and recommendations to the University of Washington Board of Regents.
- Served as a liaison between student stakeholders and IT leadership, translating survey findings into prioritized recommendations and tracking progress on action items.

Founder & President, The Programming Project

September 2022 – Present

University of Washington, Tacoma

- Founded a practical computer science organization offering weekly data structures and algorithms sessions, interview preparation, quarterly team projects, industry speaker events, and peer mentorship.
- Recruited and onboarded members each quarter and coordinated with faculty and industry professionals to organize guest speakers and project mentorship.

TECHNICAL SKILLS

Programming: Python, SQL, Java, JavaScript

Machine Learning & Retrieval: Hugging Face Transformers, TensorFlow, scikit-learn, BM25, information retrieval evaluation

Data & Research: Pandas, MIMIC-CXR, CheXpert, CheXbert, dataset curation, annotation design, error analysis

Engineering: FastAPI, Docker, Git, Jupyter, LaTeX